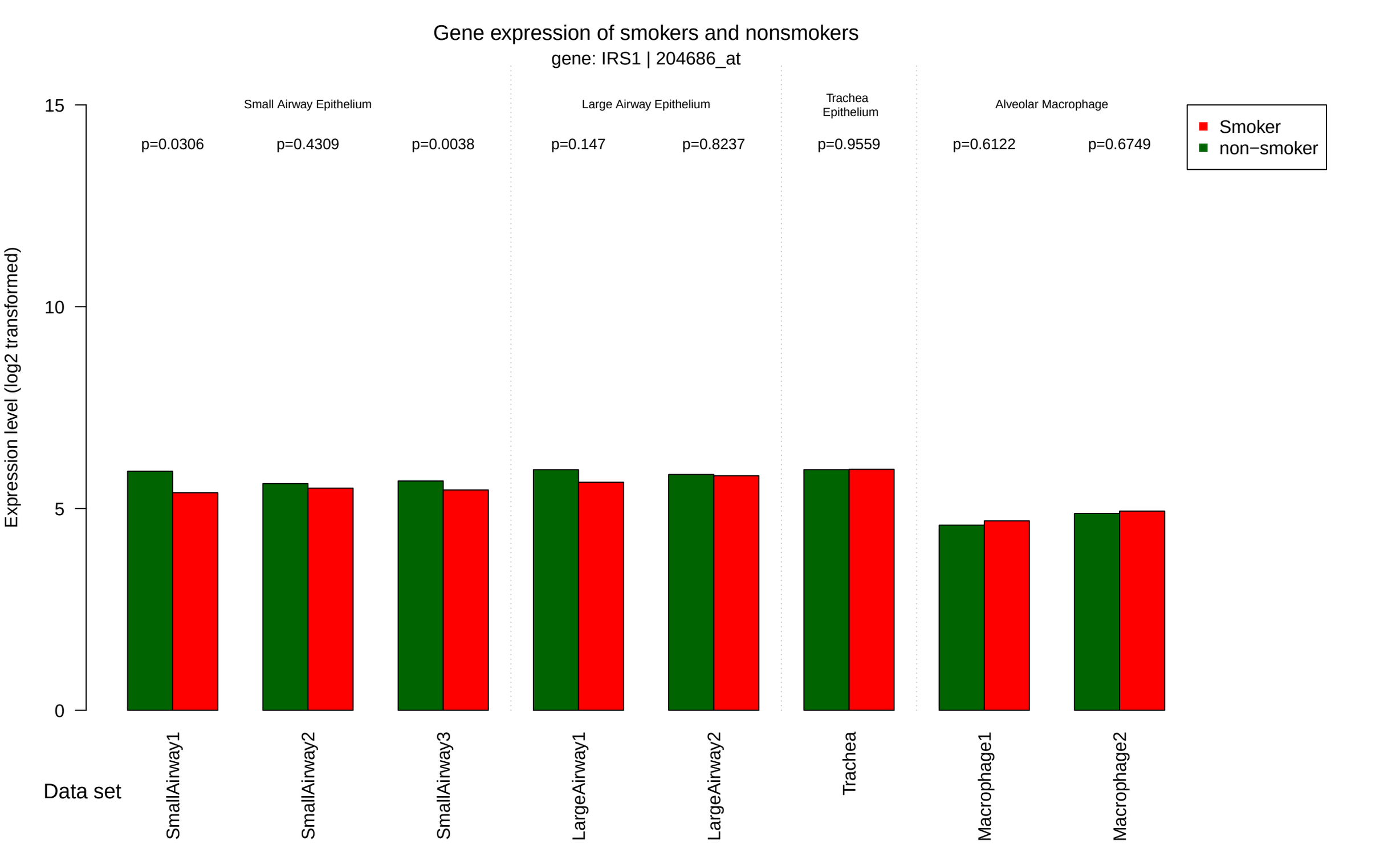
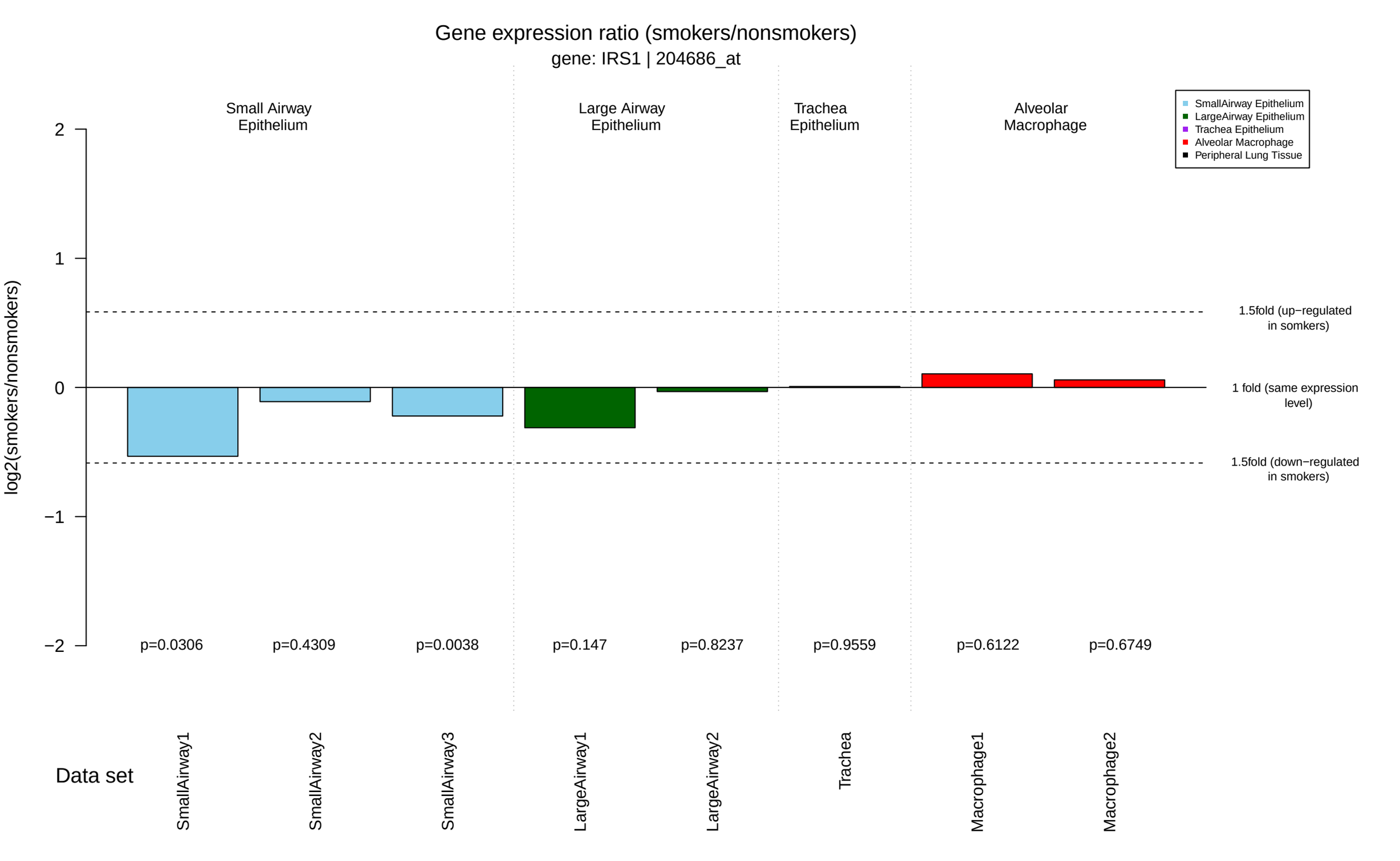
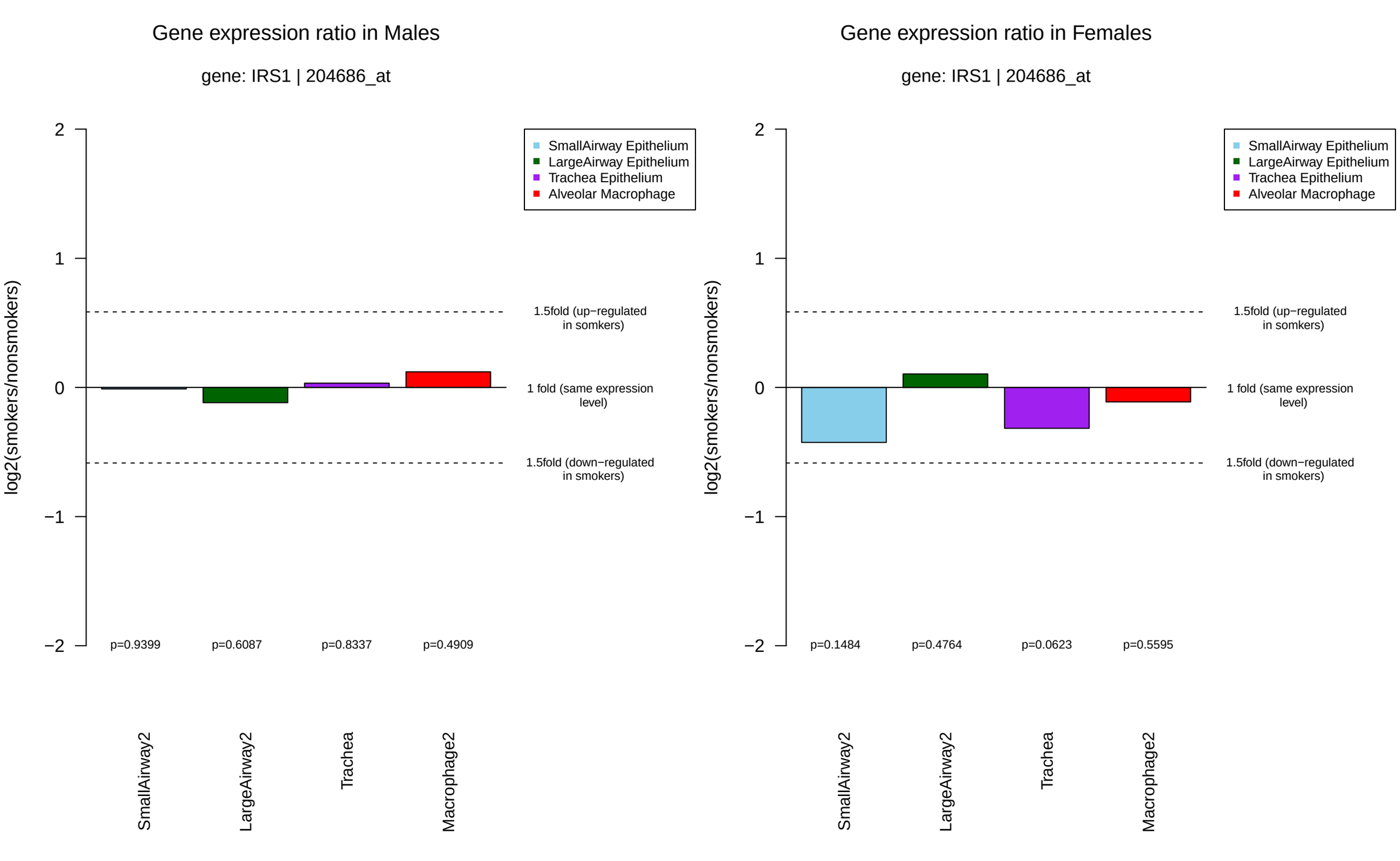
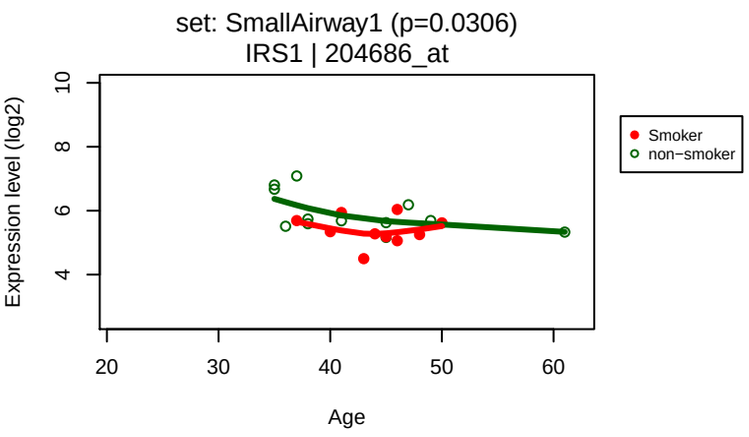
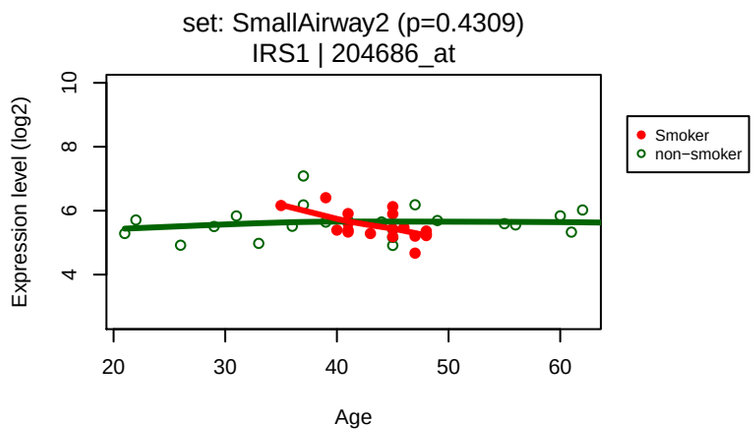
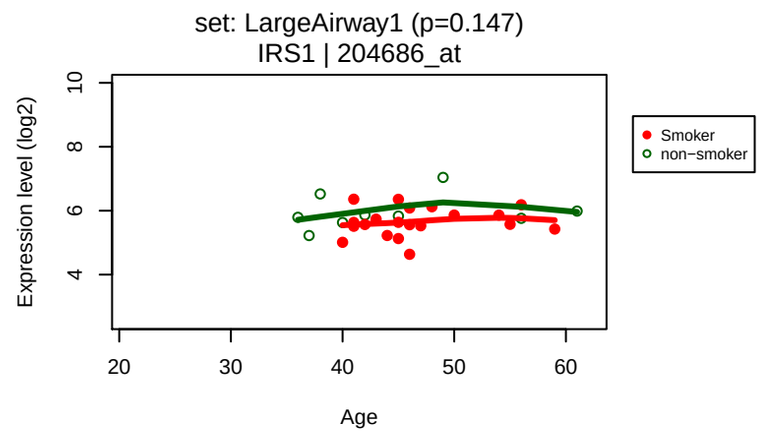

Gene expression of smokers and nonsmokers
gene: IRS1 | 204686_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0306
p=0.4309
p=0.0038
p=0.147
p=0.8237
p=0.9559
p=0.6122
p=0.6749
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

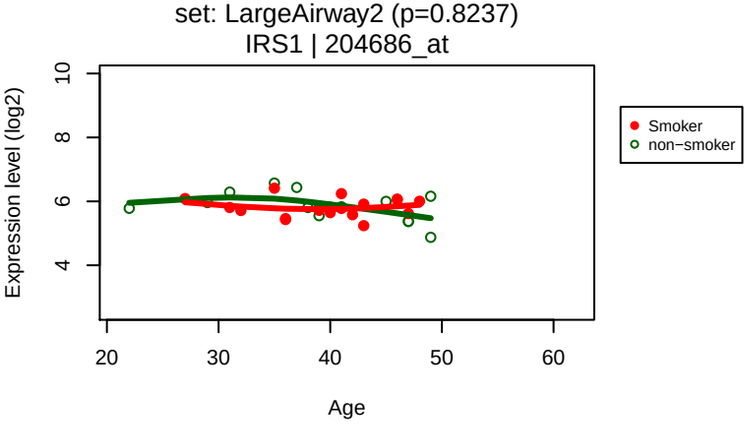

Gene expression ratio (smokers/nonsmokers)
gene: IRS1 | 204686_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0306
p=0.4309
p=0.0038
p=0.147
p=0.8237
p=0.9559
p=0.6122
p=0.6749
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

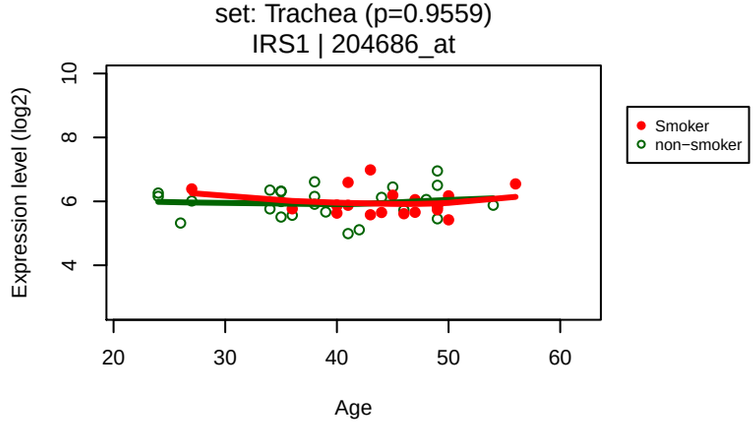

Gene expression ratio in Males
gene: IRS1 | 204686_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9399
p=0.6087
p=0.8337
p=0.4909
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRS1 | 204686_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1484
p=0.4764
p=0.0623
p=0.5595
SmallAirway2
LargeAirway2
Trachea
Macrophage2

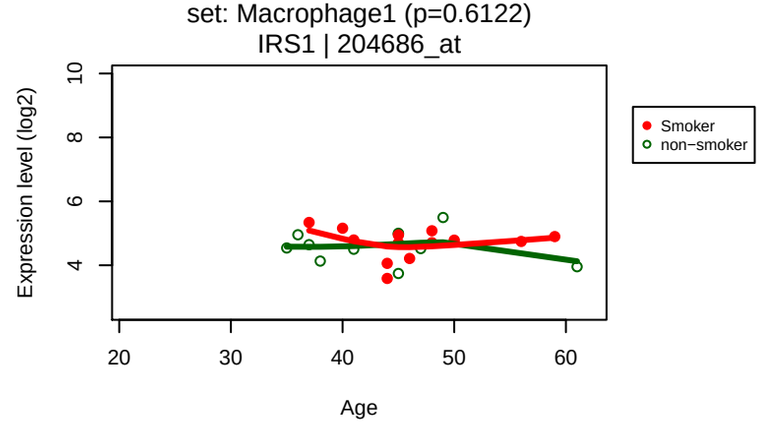

set: SmallAirway1 (p=0.0306)
IRS1 | 204686_at
Expression level (log2)
Age
Smoker
non−smoker

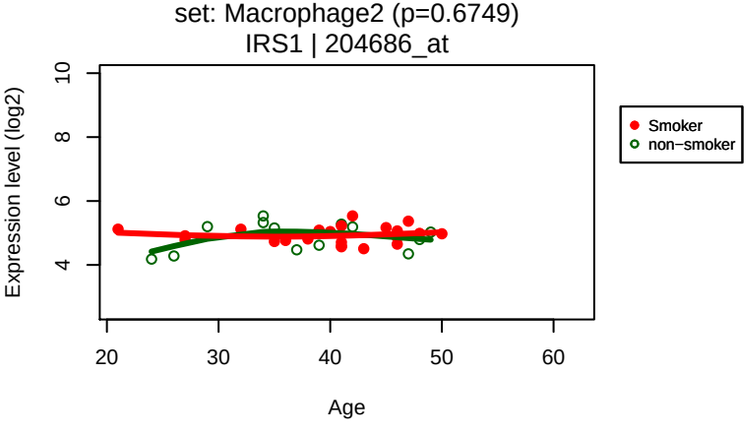

set: SmallAirway2 (p=0.4309)
IRS1 | 204686_at
Expression level (log2)
Age
Smoker
non−smoker

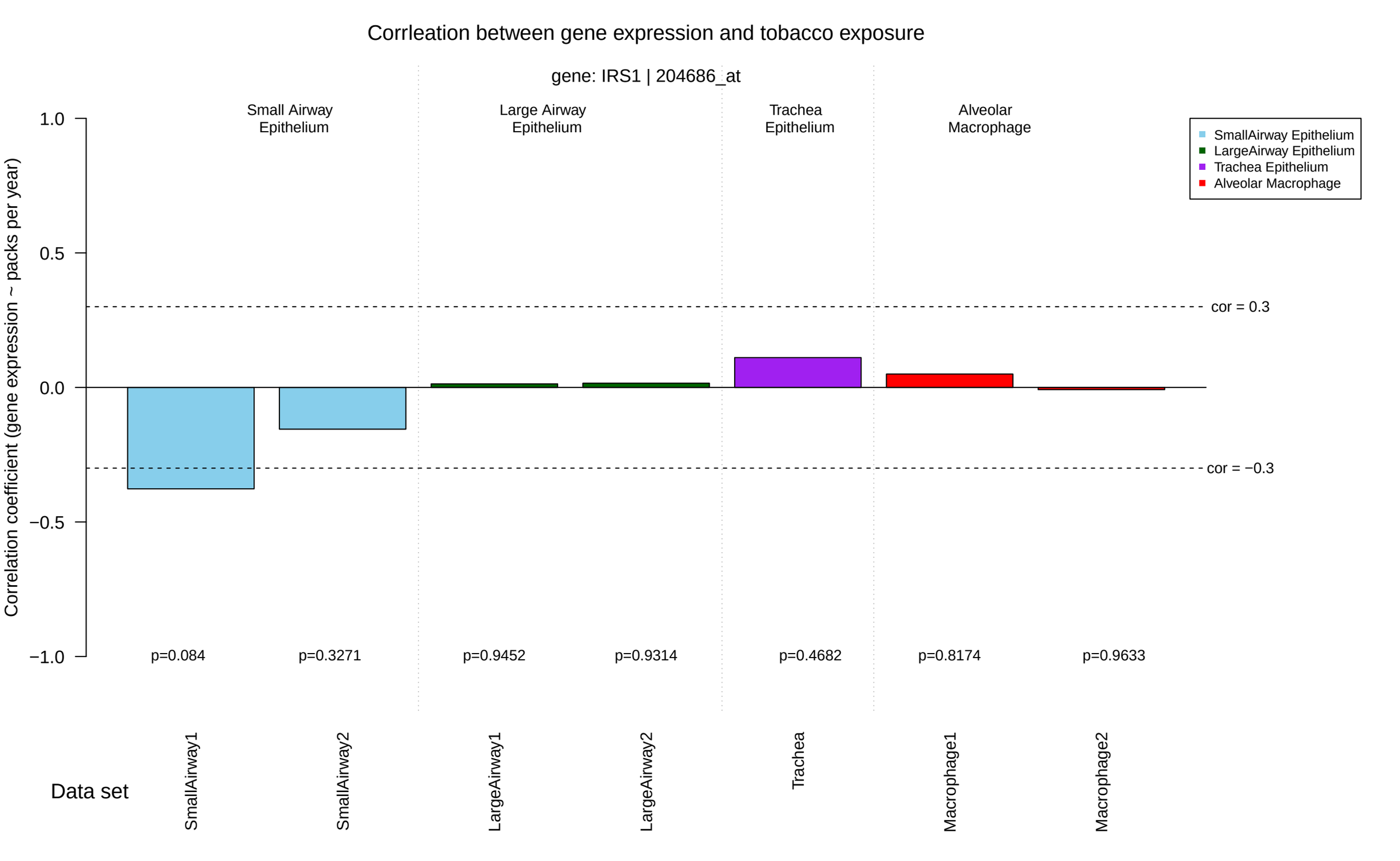

set: LargeAirway1 (p=0.147)
IRS1 | 204686_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8237)
IRS1 | 204686_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9559)
IRS1 | 204686_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6122)
IRS1 | 204686_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6749)
IRS1 | 204686_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRS1 | 204686_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.084
p=0.3271
p=0.9452
p=0.9314
p=0.4682
p=0.8174
p=0.9633
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2